1 Three-Tier Nested S-Expression Grammars and Searlean Realities

We formalize the cross-compilation of continuous pathfinding metrics, recursive PPL choice distributions, and institutional status functions within a unified three-tier LISP S-Expression syntax grammar.

1.1 The Combined Grammar Tokenization Rules

The lexical State Machine tokenizes the input alphabet by mapping sub-atomic type hierarchies across three distinct conceptual domains:

1. **Grammar 1 (Kinematic Motion):** Evaluates continuous float vectors over non-convex keep-out zones: program ::= (go vector) | (goto object) | (if boolean action action).
2. **Grammar 2 (Probabilistic Social Reasoning):** Evaluates recursive social mentalizing via level-k sampling stencils: statement $s ::= a : \text{chooses}(x \in \mathcal{X}, \text{wpp} = e) \mid a : \text{thinks}[s^*]$.
3. **Grammar 3 (Searlean Institutional World):** Extends the execution environment to include illocutionary force operators: Assertives, Directives, Commissives, Expressives, and Declarations.

1.2 The Status Function Declaration Operator

A Speech-Act declaration explicitly bypasses raw sensory ambiguity (`observes`), invoking a collective ontological inception that instantiates a new institutional property assignment directly inside the decision forest leaves:

$$\mathcal{D}_{\text{inception}}\Big(\langle C,\, X,\, Y\rangle\Big) \implies (X \text{ counts as } Y \text{ in } C) \tag{1}$$

When a misplaced branch or subject error is triggered, the DSL agent recognizes the discrepancy and back-propagates structural look-up repairs, grafting a corrected zero-uncertainty exemplar node into the living document forest branch.

1.3 Time-Bounded Deontic Logic and Legal Contract Invariants

Let $\mathcal{C}_{\text{bill}}$ be the invoice cost, $\mathcal{P}_{\text{base}}$ be the initial penalty modifier, and t_{deadline} represent the temporal boundary. The utility of evasion scales non-linearly post-deadline:

$$\mathcal{U}(a = \text{Evade}, t) = \begin{cases} 0, & \text{if } t \le t_{\text{deadline}} \\ -\mathcal{P}_{\text{base}} \cdot \gamma^{(t - t_{\text{deadline}})}, & \text{if } t > t_{\text{deadline}} \end{cases} \tag{2}$$

where $\gamma > 1$ represents the compounding interest escalation parameter. The choice vector is compiled via the exponentiated softmax calculus:

$$P(a = \text{Pay} \mid t) = \frac{\exp\left(\beta \cdot \mathcal{U}(\text{Pay}, t)\right)}{\exp\left(\beta \cdot \mathcal{U}(\text{Pay}, t)\right) + \exp\left(\beta \cdot \mathcal{U}(\text{Evade}, t)\right)} \tag{3}$$

When $t > t_{\text{deadline}}$, the non-linear penalty block forces an immediate collapse of the choice matrix, driving $\lim_{t \to \infty} P(a = \text{Pay} \mid t) = 1.0000$, unblocking the Attainable Security Performance Space V'.